Highlights

Can AI Match Professional Analysts? Evidence from a Multi-Agent System

- A multi-agent system produces equity research reports that are perceived by general-population evaluators under blinded assessment as being comparable in quality to professional reports.
- General-population participants prefer system-generated reports; finance graduate students (underpowered sample) show no detectable preference.
- LLM judges favor system-generated reports, though preferences vary substantially across model choice.

Can AI Match Professional Analysts? Evidence from a Multi-Agent System

Abstract

We evaluate whether multi-agent systems can generate equity research reports that evaluators perceive as comparable to those produced by professional analysts. Using 50 matched pairs from 2024Q3, we compare system-generated reports to Value Line using AI language model judges, finance graduate students, and general-population evaluators. Evaluation is paired and blinded, with inference based on evaluator-clustered standard errors. Blinded evaluation by general-population participants indicates that non-expert readers perceived system-generated reports as comparable in quality to professional analyst reports. The preferences among graduate students were evenly split (though this sample is underpowered), while general-population participants prefer system-generated reports. Language model judges also favor system-generated reports, though conclusions are sensitive to model choice. Portfolio tests show neither system-generated nor Value Line recommendations outperform a simple $1/N$ benchmark. Overall, the evidence suggests that multi-agent systems can produce equity research narratives that non-professional evaluators perceive as coherent, plausible, and comparable to professional reports under blinded evaluation.

Keywords: Financial Analyst, AI Agent Systems, Financial Report, Information Processing, Large Language Models

1. Introduction

The equity research industry faces a challenge. Producing analyst reports requires synthesizing corporate disclosures, earnings calls, industry dynamics, and market developments into coherent investment theses—work that is costly, capacity-constrained, and increasingly difficult to monetize. Fee pressure and regulations such as MiFID II–related research unbundling have

reduced incentives to produce sell-side research. As coverage shrinks, many firms lack professional analysis despite abundant public information.

This coverage gap has implications for market efficiency and capital allocation. Firms with limited analyst coverage exhibit greater information asymmetry and slower incorporation of firm-specific information into prices, consistent with less efficient price discovery (Hong et al., 2000). Entry-level analysts, performing data-gathering and drafting, drive research production costs. If AI systems can replicate parts of these foundational tasks at scale, the implications extend beyond technological capability to workforce structure and the economics of financial information production.

Generative AI makes scalable equity research feasible by enabling processing and synthesis of large volumes of unstructured financial text. However, single-model approaches often struggle with the breadth and internal consistency required for comprehensive analyst-style research. These limitations motivate multi-agent architectures that decompose complex research tasks across specialized components to improve reliability and coherence.

Can a multi-agent architecture produce full-length equity research comparable in quality to professional analysts' work? We address this question using a four-agent team that mirrors analyst workflows. We generate reports for 50 U.S. firms in 2024Q3 and benchmark them against matched Value Line reports.

Blinded human evaluations indicate that system-generated reports are perceived by non-expert evaluators as comparable in quality to professional reports. Graduate-level finance students show no detectable preference, though this underpowered sample is supplementary evidence. In a larger, adequately powered general-population sample—representing retail investors, who are the primary beneficiaries of expanded coverage—participants rate system-generated reports higher and prefer them more frequently. LLM judges also favor system-generated reports, though heterogeneity across models is substantial. These findings suggest that multi-agent systems can produce equity research that general-population readers perceive as coherent, plausible, and comparable in quality to professional reports. We emphasize that the powered human evidence reflects non-professional evaluators; whether professional analysts would agree remains an open question.

The investment implications are muted. Long-only portfolios based on AI recommendations do not outperform an equal-weight $1/N$ benchmark over one- and two-month horizons. This separation is informative: AI systems may support narrative synthesis without generating alpha, suggesting a po-

tential role in reducing costs and expanding coverage rather than improving stock-picking performance.

Our contribution is threefold: multi-agent systems generate reports perceived as comparable to professional analyst reports; perceived quality varies across evaluators; and comparable perceived narrative quality does not translate into abnormal investment performance. Our 2024Q3 sample provides a pre-contamination baseline before widespread analyst adoption of generative AI-intensive workflows.

2. Related Literature and Theoretical Framework

2.1. Multi-Agent Systems in Finance

Single LLMs have limitations for completing complex financial tasks. Context length constraints force selective attention across long documents (Liu et al., 2024), and without verification mechanisms, hallucinations propagate unchecked (Ji et al., 2023). Multi-agent architectures address these weaknesses through three evidence-backed mechanisms.

First, **task decomposition** reduces load. Du et al. (2024) demonstrate that multi-agent debate significantly improves factual accuracy and reasoning, reducing hallucinations by 10–30%. Second, **specialization** enables optimization. Wang et al. (2024) show that a layered Mixture-of-Agents setup, where agents condition on other agents' drafts, can outperform a single frontier model on AlpacaEval 2.0. Third, **structured coordination** ensures consistency. Han et al. (2024) compare architectures for SEC filing analysis, finding multi-agent collaboration most valuable for complex tasks like risk analysis.

In finance, systems like FinMem (Yu et al., 2025), FinCon (Yu et al., 2024), FinRobot (Yang et al., 2024), and TradingAgents (Xiao et al., 2025) deploy specialized agents for trading decisions or narrow data aggregation. However, these systems do not address the end-to-end production of analyst-style equity research reports.

2.2. AI in Financial Analysis

Evidence on LLM usefulness is mixed and sensitive to design choices. Pelster and Val (2024) show that LLMs can assist stock selection but do not reliably dominate simple benchmarks. Fieberg et al. (2024) find that LLMs can provide suitable, risk-consistent, and implementable portfolio advice, with performance comparable to leading robo-advisors and often achieving

higher Sharpe ratios. Larger foundation models show the most consistent quality. Hornuf et al. (2025) show that LLMs initially lag both human advisors and naive diversification, yet their performance improves sharply once firm-specific quantitative or qualitative information is supplied. Cao et al. (2024) demonstrate complementarity between human judgment and AI capabilities. Coleman et al. (2022) find robo-analyst portfolios outperform human analysts on return metrics.

For equity research specifically, this literature lacks systematic evaluation of AI-generated reports against professional benchmarks through designs that address LLM bias concerns.

2.3. Evaluation Methodology

Zheng et al. (2023) show LLM judges can approximate human ratings, but reliability is domain-specific. Critically, Panickssery et al. (2024) document self-preference bias: GPT-4 is 73.5% accurate at distinguishing its own text and exhibits preference for self-generated content. Li et al. (2025) find LLMs favor LLM-generated text by 10–25% in evaluation tasks. These findings motivate our dual-evaluation design combining LLM judges with blinded human assessment.

3. Data and Empirical Design

[INSERT FIGURE 1 HERE]

3.1. Multi-Agent Architecture

We implement a four-agent hierarchical team using CrewAI, where a manager agent coordinates activity. All agents use OpenAI's GPT-4o. The model's October 2023 knowledge cutoff mainly affects priors because agents retrieve firm-specific facts. Each agent is assigned a specialized role and tool set. The agents are as follows:

Research Analyst

The Research Analyst conducts qualitative analysis: business model assessment, competitive positioning, catalysts, and recent developments. Tools include web search and SEC filing parsers.

Financial Analyst

The Financial Analyst performs quantitative evaluation: statement analysis, ratio calculation (P/E, debt-to-equity, ROE), growth metrics, and peer comparisons. Its tools include the SEC filing parser, web search, and a calculator tool.

Investment Advisor

The Investment Advisor synthesizes upstream outputs into a unified thesis and Buy/Hold/Sell recommendation, relying primarily on internal team inputs instead of external tools.

Strategy Manager

The Strategy Manager coordinates workflows, enforces a five-part template, and triggers targeted revisions when detecting inconsistencies.

We developed custom tools using CrewAI's framework. The web search tool allows the agent to search a pre-defined list of conventional news sources. The SEC filings tool allows the agent to read and search the entirety of a firm's filings.

The workflow proceeds hierarchically. The Strategy Manager sequentially triggers qualitative research, filings analysis, financial analysis, and final recommendation. The task description, context, and expected output are passed to the selected agent in JSON. Each agent returns a markdown response in JSON, and the Manager reviews drafts for completeness and consistency, issuing follow-up instructions when needed.

3.2. Sample Construction

We generate reports for 50 distinct publicly traded U.S. firms sampled across major GICS sectors in 2024Q3. Inputs comprise the recent 10-K and 10-Q filings plus news. For each system-generated report, we obtain a matched Value Line report for the same firm and quarter. Value Line provides an appropriate benchmark: reports are produced by professional analysts under consistent editorial standards, enabling matched-pair comparison.

3.3. Evaluation Design

LLM Evaluation

Four heterogeneous models serve as judges: Gemini 2.5, Gemma 3, MistralNemo, and Llama 3.1. We selected models from different providers and

architectures to reduce dependence on any single model's preferences. Each judge receives an anonymized report, assigns a single 1–10 score based on five dimensions (clarity/structure, depth of analysis, data relevance, logical consistency, actionable insight), and performs a forced-choice head-to-head comparison. For score-based comparisons, we analyze within-pair differences between system-generated and Value Line reports and report paired t-tests of whether the mean difference equals zero.

Human Evaluation

We recruit two human evaluator samples: Fifteen Master of Science in Finance (MSF) students and a general-population sample recruited via the online recruitment platform Prolific. Prolific participants were not screened for finance expertise. This design choice reflects our focus on general-population evaluation: equity research is consumed not only by professionals but also by retail investors who may lack formal finance training. For this population, perceived report quality—including clarity, logical structure, and actionable conclusions—constitutes economically relevant evidence on how expanded coverage would be received. Professional analysts may weight quality dimensions differently, so whether system-generated reports satisfy expert standards remains open.

[INSERT FIGURE 2 HERE]

Evaluators complete the same tasks for matched report pairs under a fully blinded protocol: source labels removed, presentation order randomized, and pair sequence randomized across evaluators. Both reports were presented in plain-text format within identical interface containers to minimize visual formatting differences. However, structural elements such as section headers and paragraph organization were preserved from the original sources. To the extent that multi-agent systems produce more clearly organized prose, this organizational quality is reflected in evaluator ratings. We interpret this as a genuine dimension of report quality for non-expert readers rather than a confound, consistent with evidence that processing fluency influences how investors respond to disclosure readability (Rennekamp, 2012).

Humans use the same rubrics and head-to-head preference questions as the LLM judges. Forty-nine participants recruited via Prolific each evaluate four matched report pairs. This yielded 196 paired evaluations; two

were excluded for errors, leaving 194. Each MSF student evaluates a randomized subset of three report pairs. For each evaluated pair, we compute the within-pair score difference between system-generated and Value Line reports. Statistical inference is based on tests of whether the mean paired difference equals zero, with standard errors clustered at the evaluator level to account for repeated evaluations by the same participant. The Prolific sample provides adequate power for effects of the observed magnitude. The MSF student sample alone is underpowered and unsuitable to detect small effects, and its results should be noted as supplementary evidence.

3.4. Portfolio Evaluation

Following Hornuf et al. (2025), we evaluate performance relative to an equal-weight $1/N$ benchmark that holds all sample firms and is rebalanced at the same frequency as our test portfolios. We then construct long-only "Buy" portfolios for each strategy. For Value Line, we convert their five-point technical rating into a binary signal and treat stocks with a technical rating of 1 or 2 as "Buy" (included in the long portfolio), while ratings of 3–5 are treated as "Hold/Sell" (excluded). We compute buy-and-hold returns over one- and two-month horizons and estimate Fama–French six-factor regressions for the multi-agent system, Value Line, and $1/N$ benchmark portfolios.

4. Results

4.1. Report Quality: Comparable Under Human Evaluation

Table 1 summarizes sample characteristics and evaluation outcomes. Under blinded evaluation, system-generated reports are rated comparably to professional analyst reports overall, with the strongest preference emerging among general-population evaluators.

[INSERT TABLE 1 HERE]

The human evaluators provide complementary evidence from two samples. Prolific evaluators assign a mean quality score of 8.129 to the multi-agent system versus 6.644 to Value Line (difference 1.485, $p < 0.01$) and prefer the system-generated report 76.8% of the time. MSF students (though underpowered) show no detectable difference in mean perceived quality; their

head-to-head choices favor the system-generated report 51.1% of the time, statistically indistinguishable from an even split. We treat the MSF evidence as supplementary given limited power to detect modest effects.

LLM judges also favor the multi-agent system, assigning a mean score of 7.079 versus 6.807 for Value Line (difference 0.272, $p < 0.01$) and preferring system-generated reports 71.0% of the time. The results indicate perceived quality comparable to professional reports across evaluator types.

The stronger preference in the Prolific sample is consistent with how non-expert investors consume research, emphasizing clear narratives, salient risks, and actionable conclusions (Rennekamp, 2012). Because these dimensions may not align perfectly with professional analyst priorities, we emphasize the conservative inference supported across samples: system-generated reports are perceived as comparable in quality to professional reports under blinded evaluation.

[INSERT TABLE 2 HERE]

Table 2 shows substantial heterogeneity across LLM judges, with preferences ranging from 48% (Gemini 2.5) to 96% (Gemma 3). This dispersion reinforces why human evaluation is essential: LLM-as-judge conclusions are unstable across models.

4.2. Cross-Sectional Heterogeneity

Table 3 splits firms by analyst coverage. In the MSF sample, evaluators show a larger (though not statistically significant) gap favoring the multi-agent system among below-median coverage firms (gap 1.136, $p = 0.103$) compared to above-median coverage firms (gap 0.130, $p = 0.790$). In the Prolific sample, the gap is positive and statistically significant in both coverage groups, and remains larger for below-median coverage firms (gap 1.833, $p < 0.01$) than for above-median coverage firms (gap 1.098, $p < 0.01$). LLM judges do not show a stable pattern. The cross-sectional pattern suggests automated synthesis is potentially useful in information-sparse environments where analyst attention is limited.

[INSERT TABLE 3 HERE]

4.3. Portfolio Performance and Economic Value

Table 4 evaluates investment implications. Over one month, the $1/N$ benchmark earns 2.89% versus 1.87% for the multi-agent system and 1.47% for Value Line. Over two months, returns are similar (1.43%, 1.63%, and 1.40%, respectively). Sharpe ratios show the same pattern (Table 4), and none of the strategy-by-benchmark differences are statistically significant.

Fama–French six-factor alphas are statistically insignificant across all strategies. Neither AI nor human analyst recommendations generate abnormal returns relative to diversification.

[INSERT TABLE 4 HERE]

This null result is informative. It suggests that AI can produce analyst-style reports that evaluators rate favorably without generating alpha, consistent with the separation predicted if markets efficiently incorporate public information. The value proposition of AI equity research may lie more in cost reduction and coverage expansion than in incremental alpha generation.

5. Conclusion

System-generated reports are perceived by non-expert evaluators as comparable in quality to professional equity research under blinded evaluation. Preferences among MSF students are split roughly evenly, assigning statistically indistinguishable quality scores across system-generated and professional reports, though this sample is underpowered. In the larger Prolific sample, evaluators assign higher average scores to the multi-agent system's reports and prefer them more frequently. The human evidence indicates that multi-agent systems can generate analyst-style research that non-expert evaluators perceive as coherent, plausible, and comparable to professional reports. LLM judges also favor system-generated reports, but preferences vary across models.

Portfolio tests show that neither the multi-agent system nor Value Line outperforms naive diversification. These results suggest that the value of multi-agent AI may lie more in supporting narrative synthesis and broader coverage than in generating alpha.

These findings bear on drafting and synthesis within the analyst workflow. AI appears capable of supporting these tasks in ways that non-expert

evaluators perceive as coherent and plausible, potentially supporting broader coverage while leaving domain-specific judgment to human analysts. Our 2024Q3 setting provides a baseline for tracking how report quality evolves as AI adoption accelerates.

Our human evidence comes from general-population participants who were not screened for finance expertise. While this design is appropriate for assessing reports' value to retail investors, it does not establish professional-level, factual, or domain-specific quality, nor whether professional analysts would perceive equivalent quality. Future research should examine expert evaluation and downstream decision-making outcomes.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the authors used OpenAI's ChatGPT to assist with language refinement and clarity improvements. All content produced with the assistance of these tools was reviewed, verified, and edited by the authors, who take full responsibility for the final manuscript and its conclusions.

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Tables

Table 1: Sample Characteristics and Evaluation Outcomes

Panel A: Firm characteristics (Firm level, $N = 50$)

Variable	Mean	Median	SD	P25	P75
Market cap (USD mn)	522.00	247.60	772.60	159.30	469.00
Log(Market cap, USD)	19.46	19.33	1.05	18.89	20.00
Analyst coverage	19.68	21.00	16.17	0.00	30.75

Panel B: Evaluation outcomes (Evaluator–Firm level)

	Agentic Team	Value Line	Diff	t-stat	N
LLM Evaluators					
Quality score	7.079	6.807	0.272	3.17***	200
Preference for Agentic Team (0–1)	0.710	–	–	–	200
Human Evaluators: Master's students					
Quality score	7.088	6.466	0.622	1.17	45
Preference for Agentic Team (0–1)	0.511	–	–	–	45
Human Evaluators: General-population (Prolific)					
Quality score	8.129	6.644	1.485	5.43***	194
Preference for Agentic Team (0–1)	0.768	–	–	–	194

Notes: Panel A reports firm-level characteristics. Panel B reports evaluation outcomes for reports generated by the agentic team and Value Line. Master's students represent an expert-adjacent evaluator group, while general-population participants are recruited from an online panel. Differences are computed as agentic team minus Value Line. For human samples, t-statistics are based on tests of whether the mean paired difference equals zero with standard errors clustered at the evaluator level. Results are robust to two-way clustering by evaluator and firm-pair. Significance levels: *** 1%, ** 5%, * 10%.

Table 2: Evaluator-Level Comparison of Report Quality and Preferences

Panel A: Quality score comparison (Report level)

		Agentic Team	Value Line		
Evaluator	N	Mean Score	Mean Score	t-stat	p-value
LLM Evaluators					
Gemini 2.5	50	6.640	6.520	0.63	0.528
Gemma 3	50	6.560	6.260	2.41	0.018**
MistralNemo	50	7.228	6.986	2.03	0.045**
Llama 3.1	50	7.888	7.460	3.63	< 0.01***
Human Evaluators					
Master's students	45	7.088	6.466	1.17	0.243
Prolific (general pop.)	194	8.129	6.644	5.43	< 0.01***

Panel B: Head-to-head preference rates

Evaluator	N	Agentic Team Count	Value Line Count	Preference for Agentic	p-value (Test)
LLM Evaluators					
Gemini 2.5	50	24	26	48%	0.888
Gemma 3	50	48	2	96%	< 0.01***
MistralNemo	50	28	22	56%	0.480
Llama 3.1	50	41	9	82%	< 0.01***
Human Evaluators					
Master's students	45	23	22	51%	0.888
Prolific (general pop.)	194	149	45	77%	< 0.01***

Notes: Panel A reports mean quality scores for agentic team versus Value Line by evaluator type. Panel B reports the frequency with which the evaluator preferred the agentic team report in head-to-head comparisons. For human samples, p-values are based on tests of whether the mean paired difference (Panel A) equals zero and whether the mean preference rate (Panel B) equals 0.5, with standard errors clustered at the evaluator level. Results are robust to two-way clustering by evaluator and firm-pair. Master's students represent an expert-adjacent evaluator group; general-population participants are recruited from an online panel. For readability, very small p-values are reported as < 0.001. Significance levels: *** 1%, ** 5%, * 10%.

Table 3: Cross-Sectional Heterogeneity by Analyst Coverage

Group	N_{Firms}	N_{Obs}	Agentic Team	Value Line	Gap	p-value
AI Evaluators						
Above-median coverage	24	96	7.200	6.786	0.414	$< 0.001^{***}$
Below-median coverage	26	104	6.967	6.825	0.142	0.168
Human Evaluators: Master's students						
Above-median coverage	23	23	6.826	6.696	0.130	0.790
Below-median coverage	22	22	7.364	6.227	1.136	0.103
Human Evaluators: General-population (Prolific)						
Above-median coverage	24	92	7.913	6.815	1.098	$< 0.001^{***}$
Below-median coverage	26	102	8.324	6.490	1.833	$< 0.001^{***}$

Notes: This table reports mean report quality by analyst coverage group (above- versus below-median) and evaluator type. "Gap" is the mean difference (agentic team minus Value Line). For human samples, t-statistics are based on tests of whether the mean paired difference equals zero with standard errors clustered at the evaluator level. Results are robust to two-way clustering by evaluator and firm-pair. For readability, very small p-values are reported as < 0.001. Significance levels: *** 1%, ** 5%, * 10%.

Table 4: Portfolio Performance and Factor Regressions

Panel A: Portfolio performance vs. 1/N benchmark

Strategy	Mean Excess Return (%)	Annualized Sharpe Ratio	t-stat vs. 1/N (t-stat)
1-month horizon			
1/N Benchmark	2.89	3.06	–
Agentic Team	1.87	1.49	−0.47
Value Line	1.47	1.44	−1.09
2-month horizon			
1/N Benchmark	1.43	1.54	–
Agentic Team	1.63	1.39	0.19
Value Line	1.40	1.46	−0.03

Panel B: Fama–French 6-factor regression results

	Mth.	Factor Loadings					
Strategy	α (%)	Mkt	SMB	HML	RMW	CMA	MOM
1-month horizon							
1/N Benchmark	−0.32	0.77***	−0.05	−0.02	−0.07	−0.13	−0.17
	(−0.33)	(10.13)	(−0.52)	(−0.29)	(−0.36)	(−0.82)	(−1.52)
Agentic Team	−1.67	0.95***	0.09	−0.16	0.36	0.01	0.20
	(−1.12)	(10.70)	(0.85)	(−2.44)	(1.89)	(0.06)	(1.65)
Value Line	−1.66	0.75***	0.02	−0.23	0.29	−0.38	0.03
	(−1.22)	(5.44)	(0.11)	(−2.26)	(0.95)	(−1.30)	(0.14)
2-month horizon							
1/N Benchmark	−0.55	0.84***	−0.07	0.02	−0.04	−0.01	−0.18***
	(−0.87)	(18.22)	(−1.25)	(0.43)	(−0.55)	(−0.18)	(−3.19)
Agentic Team	−0.66	0.96***	−0.00	−0.11	0.15	0.18	0.12
	(−0.69)	(14.64)	(−0.04)	(−2.45)	(1.55)	(1.70)	(1.68)
Value Line	−1.08	0.88***	−0.08	−0.15	0.25	−0.11	−0.18
	(−1.06)	(12.34)	(−0.77)	(−2.64)	(2.42)	(−0.98)	(−1.99)

Notes: Panel A reports monthly-equivalent mean excess returns (in percent) and annualized Sharpe ratios computed from daily excess returns within each horizon window. The "t-stat vs. 1/N" column reports the Newey–West HAC t-statistic for the null that the mean daily return difference between the strategy and the 1/N benchmark is zero, i.e., $\mathbb{E}[r_{s,t} - r_{1/N,t}] = 0$. Panel B reports factor loadings from regressions of daily portfolio excess returns on the Fama–French five factors plus momentum (UMD); α is reported as a monthly percent equivalent (daily intercept scaled by 21 trading days). t-statistics are in parentheses beneath each coefficient and use Newey–West HAC standard errors (lag 0 for the 1-month window and lag 1 for the 2-month window). Significance levels: *** 1%, ** 5%, * 10%.

Figures

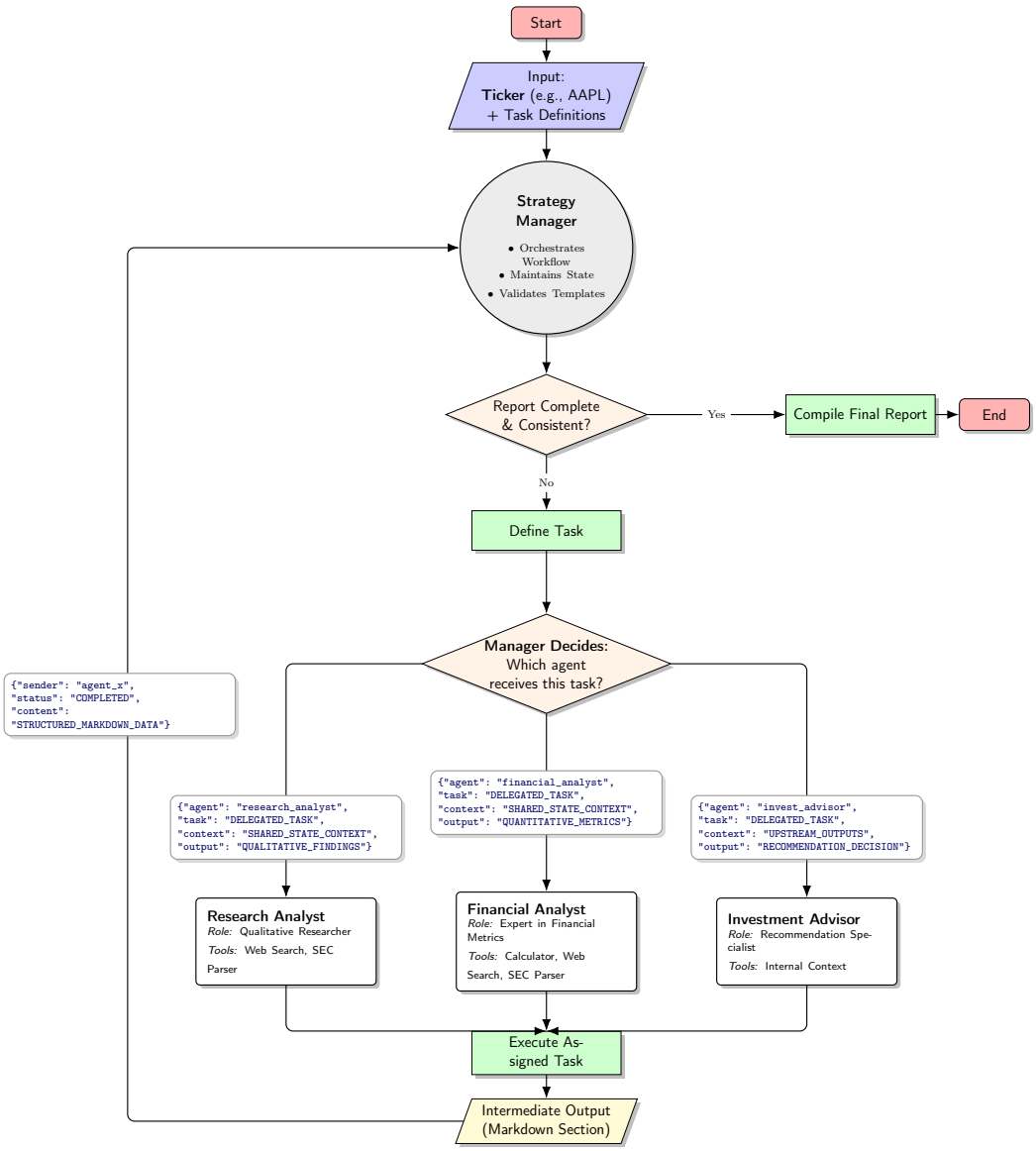


Figure 1: Multi-agent architecture and workflow. The **Strategy Manager** serves as the central controller, orchestrating a four-step workflow by assigning abstract tasks via structured JSON to specialized agents. The **Research Analyst** (Qualitative & Filings), **Financial Analyst** (Quantitative), and **Investment Advisor** (Recommendation) execute tasks using designated tools and return structured markdown findings to the shared state. A revision loop ensures completeness before the final report is compiled.



Figure 2: **Example pair shown to evaluators.** The figure displays an example of the reports as they appeared to participants in the blinded evaluation interface. In this example, Report A is the system-generated report and Report B is the Value Line report. The reports were shown with their original formatting preserved, and the assignment of reports to the A/B labels as well as their presentation order was randomized across participants.